Multimodal RAG for Advanced Chemistry: Targeted Information Retrieval and Question Answering from Slide-Based Lectures

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Abstract

We present a multimodal Retrieval-Augmented Generation (RAG) system for advanced organic chemistry education, designed to retrieve relevant lecture slides and answer domain-specific questions. Using two benchmarks of multiple-choice questions grounded in real slide-caption pairs, we evaluate three large multimodal models: Qwen, Gemini Flash, and Gemini Pro, across no-context, contextual, and RAG-based settings. This project highlights the potential of multimodal models for specialized scientific reasoning and as educational tools in chemistry.

1 Introduction

In recent years, the intersection of large multimodal models and domain-specific knowledge extraction has emerged as a promising avenue for advancing specialized fields such as chemistry. This project addresses the challenge of developing a multimodal Retrieval-Augmented Generation (RAG) system tailored to advanced organic chemistry education. The primary aim is to enable more interactive, accurate, and accessible knowledge retrieval for complex lecture content in advanced chemistry.

2 Dataset

The project builds upon the Synthesis Workshop dataset, a unique and comprehensive collection from which we took 174 advanced chemistry lectures. The dataset presents a rich multimodal resource that includes slide images, textual content, molecular structures (e.g., SMILES), and mechanistic descriptions. The scope of the project evolved to building a robust question-answering system capable of providing context-aware answers to questions about specific slides within the entire lecture corpus. The developed system leverages state-of-the-art visual language models to encode and retrieve slide- specific information. Using a Retrieval-Augmented Generation pipeline, the system first locates relevant slides and extracts accompanying information, then generates informed answers to MCQs. To ensure practical relevance, an interactive interface has been integrated, allowing users to query the model, retrieve targeted explanations, and directly navigate to relevant timestamps in associated lecture videos hosted on the Synthesis Workshop YouTube channel. This bridges static slide information with dynamic lecture content, enriching the learning experience.

To build the dataset of (slide, transcription) pairs, we designed a pipeline that transforms recorded lecture videos (.mp4) into aligned image-text pairs. First, videos were sampled at a rate of 1 frame per second to extract slide frames, using a pixel-wise mean difference threshold of 1.2 to detect slide changes while ignoring minor animations. A slide transition buffer of 1.5 seconds was applied

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to avoid capturing blurred frames. Each detected slide was saved as a .jpg image, along with a timestamp. Transcriptions were generated using Whisper, which provides sentence-level timestamps. These timestamps were then aligned with the extracted slide times to assign each sentence to the appropriate slide. A 1-second lag parameter was used to account for early speech about upcoming slides. This process yielded a clean dataset of paired slide images and corresponding transcribed text files. Please find the detailed information and instructions on how to run the pipeline for future reference (e.g to update the data) on the repository.

3 Retrieval

The Retrieval-Augmented Generation (RAG) pipeline used in this work is divided into two components: retrieval and generation. In this section, we focus on designing a retrieval implementation adapted to the dataset and the task at hand. We aim to implement a retrieval designed for a multimodal task, which implies a hybrid retrieval that effectively leverages both image and text data to answer domain-specific questions.

3.1 Methods

This section details the methods developed for the retrieval component, tailored specifically to our multimodal chemistry lecture dataset.

We progressively designed and evaluated a series of hybrid retrieval methods, each improving upon its predecessor by addressing shortcomings and refining component choices. We aimed at minimizing the number of retrieved slides while still achieving a good retrieval accuracy. Please note that the pool of slides is of size 1185.

Method 1: SentenceTransformer + CLIP This baseline approach used a dual encoder architecture to incorporate both text and visual information.

Text Encoding: Slide captions and user queries were encoded using `all-MiniLM-L6-v2` from SentenceTransformer, with embeddings indexed via FAISS.

Image Encoding: Slide images were encoded using CLIP (ViT-Large-Patch14), and user queries were also processed through CLIP's text encoder to project them into the visual-latent space.

Hybrid Scoring: Scores were combined using a weighted sum:

$$S_{hybrid} = (1 - w) \cdot S_{text} + w \cdot S_{image}.$$

Neighborhood expansion was applied to include contextually adjacent slides. This method achieved 68.5% accuracy with $k = 3$, $w = 0.8$, and three neighbors per side.

Method 2: BGE + CLIP To improve semantic matching, the text encoder was upgraded to `BAAI/bge-large-en-v1.5`.

Text Encoding: All slide captions and queries were embedded using BGE and indexed with FAISS.

Image Encoding: Slide images continued to be encoded with CLIP.

Results: Accuracy improved to 71.2% with the same parameters as Method 1. Tuning w to 0.5 and reducing neighbors led to 77.8% accuracy with fewer slides retrieved (e.g., 7.8 with one neighbor).

Method 3: BGE + Tesseract OCR To address CLIP's limitations in capturing chemical notation, this method replaced image embeddings with text extracted via Tesseract OCR.

Workflow: Tesseract extracted raw text from slides, which was then encoded using BGE. Retrieval used a hybrid score:

$$S_{hybrid} = (1 - w) \cdot S_{caption} + w \cdot S_{OCR}.$$

Results: Achieved 77.8% accuracy with minimal neighborhood expansion (1 per side) and retrieved an average of 7.2 slides.

Method 4: BGE + EasyOCR Building on Method 3, Tesseract was replaced with EasyOCR for improved layout handling and text detection.

Workflow: EasyOCR processed slides into aggregated line text, which was embedded using BGE.

Results: Achieved 94.4% accuracy at $k = 5$, $w = 0.5$, and three neighbors per side, retrieving around 20 slides per query.

Method 5: BGE + DONUT Vision-Language OCR This final method evaluated DONUT, a transformer-based image-to-text model fine-tuned with chemical prompts.

Workflow: Each slide was processed using DONUT with a custom prompt, and the extracted text was embedded with BGE.

Results: Preliminary experiments yielded 74.1% accuracy at $k = 6$, $w = 0.5$, and three neighbors, with 28 slides retrieved per query. The performance of method 5 could be further improved with prompt tuning.

3.2 Results

The comparative results are showcased in 1 and highlight the progressive improvement of retrieval accuracy. The use of stronger text encoders and more sophisticated OCR methods significantly boosted performance. In particular, EasyOCR exposed chemical structure information absent in captions, indicating the potential of domain-specific vision-language retrieval.

Table 1: Comparative accuracy and average slides retrieved for each method.

Method	Accuracy (%)	Avg. Slides Retrieved
SentenceTransformer + CLIP (Method 1)	68.5	14.6
BGE + CLIP (Method 2)	71.2 – 77.8	12.1 – 15.0
BGE + Tesseract OCR (Method 3)	74.1 – 77.8	6.6 – 7.2
BGE + EasyOCR (Method 4)	83.3 – 96.3	10.3 – 24.0
BGE + DONUT (Method 5)	74.1	28.0

4 Generation Benchmark

4.1 Hard Benchmark

A key contribution of this work is the design and implementation of a domain-specific benchmark for evaluating the system's performance. The 54 multiple choice questions were curated by Zlatko Joncev. They test the model's ability to retrieve and reason over highly technical chemical concepts across lectures. By experimenting with various retrieval strategies and generation methods, the project aims to identify the trade-offs and capabilities of current multimodal language models in supporting complex, domain-specific education tasks. Examples of the MCQs included in the benchmark are given in 1.

4.1.1 Methods

We evaluate the performance of three state-of-the-art multimodal models: Qwen VL-2.5, Gemini Flash 1.5, and Gemini Pro 1.5. Each model is prompted with a slide-caption pair retrieved using our RAG system with optimized parameters:

- `top_k=6`,
- `hybrid_weight=0.5`,
- `min_score_threshold=0.3`,
- `neighbor_window=1`

The choice of hyperparameters reflects a tradeoff between retrieval accuracy and the number of slides retrieved, which must be limited to reduce contextual noise introduced to the model. We evaluated

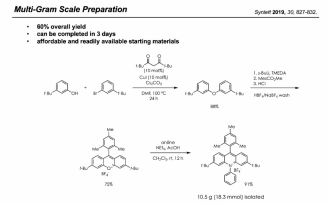
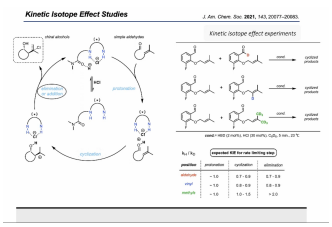
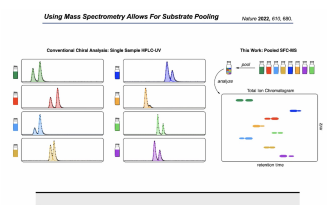
Slide	Question	Options	Ans.
	In how many chemical steps is acridinium salt synthesized? Starting material and a product of the step are separated by arrow.	a) One b) Two c) Three d) Four	c)
	Where is the molecule that starts the catalytic cycle?	a) In the middle of the cycle b) At the top left of the cycle c) At the top right of the cycle d) It is not shown	a)
	When using mass spectrometry for substrate pooling, what are the two axes in the Pooled SFC-MS method?	a) Retention time and UV activity b) Retention time and m/z c) UV activity and m/z d) Retention time and UV activity	b)

Figure 1: Multiple-choice questions aligned with chemical slide content. Each question relies on visual slide context for optimal answering.

three prompting strategies: a simple direct question, a chain-of-thought (CoT) reasoning prompt, and a prompt encouraging answer elimination. Among these, the simple prompt consistently achieved the highest performance.

Setting	Prompt
With Slide Context	"You are a chemistry expert. Answer the question based on your knowledge and the slide image." SLIDE: <image> QUESTION: question a) option1 b) option2 c) option3 d) option4 Answer: <letter> Explanation: <brief reasoning>
Without Slide Context	"You are a chemistry expert. Answer the question based on your knowledge." QUESTION: question a) option1 b) option2 c) option3 d) option4 Answer: <letter> Explanation: <brief reasoning>

Table 2: Prompt templates used with and without slide context.

4.1.2 Results

To evaluate the ability of the models to understand chemical structures and extract relevant detailed information from the images, we first evaluated their performance in answering the questions without any slide context. Given that each question has 4 options, a score of 25% corresponds to random guessing. Then, we provide the models with access to the correct slide without additional noise. This simulates a well-structured, focused input scenario, and provides an upper bound for our RAG accuracy. The result of these two experiments are shown in 3.

Model	Without Context (%)	With Context (%)	Improvement (%)
qwen	33.33 ± 6.74	42.59 ± 0.93	+8.27
gemini_flash	35.19 ± 0.00	43.52 ± 0.92	+8.33
gemini_pro	29.63 ± 2.69	49.07 ± 0.93	+19.44

Table 3: Comparison of Model Accuracy With and Without the True Slide Context

These results indicate that even without visual or textual context, the models perform above chance level. Gemini Flash leads in both accuracy and stability (zero variance). Qwen also performs consistently above random, while Gemini Pro underperforms slightly, possibly due to its more cautious or generalist decoding behavior. However, none of the models approach reliable understanding without contextual input, which highlights the necessity of including relevant slide information.

As we can see, there is an improvement in accuracy for all models once relevant slide and caption context is included. This demonstrates that the models benefit from visual input and are able to extract some chemical reasoning or visual information from the slides. Gemini Pro remains the top performer, with a close to 20% improvement, while Gemini Flash and Qwen benefit from around 8% improvement in accuracy.

These results suggest that while all three models are capable of basic visual-textual integration, their ability to interpret domain-specific chemical slides is still limited, and likely influenced by how well their vision encoder aligns with fine-grained scientific diagram features.

Now with the full RAG pipeline, we attempt to emulate a more realistic information retrieval setting. We test the impact of limiting the number of retrieved data points to 4 pairs of slide and caption. The results are shown in 4.

Model	Mean Generation Accuracy	Mean Response Time (s)
Qwen	41.6%	4.59
Gemini Flash	35.19%	2.39
Gemini Pro	44.44%	3.69

Table 4: Performance averaged across two benchmark runs with a limit of 4 retrieved slides yielding 60.00% retrieval accuracy.

By limiting the retrieved context to 4 slides, we ensure a tighter, more focused context window that may help the model stay on target. However, this comes with the risk of missing the true slide if the retriever fails.

As expected, the performance of this experiment falls between that of the models given no context and given the true slide. We observe that between all 3 models, Gemini Flash has a lower tolerance to extra noise.

The results are plotted in 2.

4.2 Synthetic Benchmark

To complement the original challenging dataset, we constructed a new synthetic benchmark tailored for controlled evaluation. This benchmark consists of 20 multiple-choice questions, each paired with a chemistry lecture slide.

4.2.1 Methods

The questions were generated using GPT-4o and are designed to be difficult to answer without the corresponding visual context, yet significantly easier when the relevant image is available. This setup allows for a systematic comparison of retrieval-augmented models, specifically evaluating their ability to reason over visual scientific content. Example questions are shown in 3.

The results reveal an interesting tradeoff between a model's prior chemistry knowledge and its ability to benefit from visual context. Both Gemini Pro and Gemini Flash show modest performance gains when provided with the true slide context, suggesting that these models already possess substantial

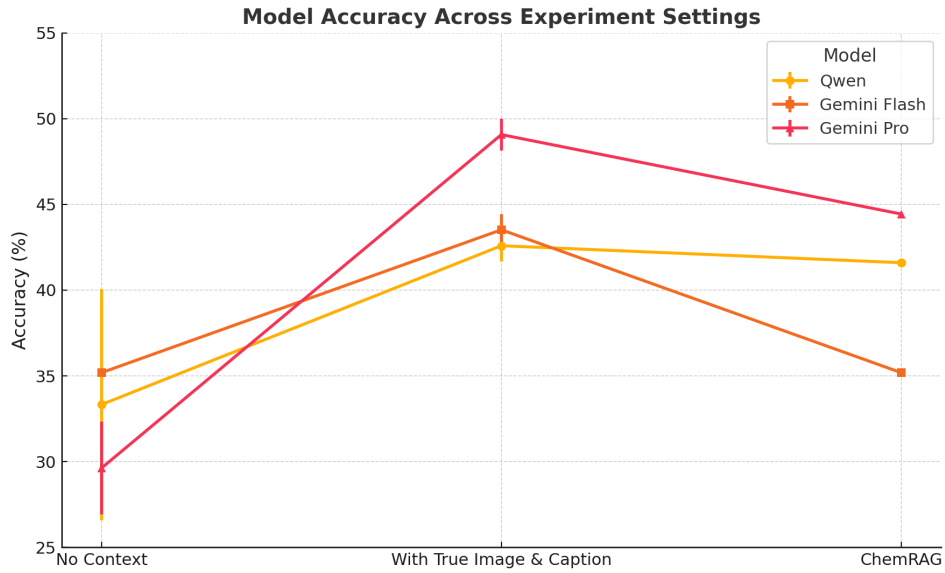


Figure 2: Accuracy and standard deviation comparison of three multimodal models (Qwen, Gemini Flash, Gemini Pro) across experimental conditions: no context, with true image and caption, and ChemRAG.

Table 5: Comparison of model performance with and without the true slide context on synthetic data, averaged over 3 runs

Model	Without Context (%)	With Context (%)	Improvement (%)
Qwen	60.00 ± 4.08	71.67 ± 2.36	+11.67
Gemini Flash	70.00 ± 3.21	75.00 ± 4.08	+5.00
Gemini Pro	68.33 ± 2.36	75.00 ± 4.08	+6.67

domain knowledge and can answer many questions even without visual support. In contrast, Qwen exhibits the largest improvement, indicating that it relies more heavily on the visual information to compensate for a weaker internal understanding of chemistry. This highlights a key distinction: models with less pre-existing knowledge may benefit more from image retrieval, whereas stronger models gain less because they already infer much from the question alone.

Running the entire RAG pipeline on the synthetic data, with a retrieval limited to 4 slides, yielding a 60% retrieval accuracy, we get the results shown in 6. As can be observed, the performance falls between the accuracy with and without context. Moreover, the generation is stable as the standard deviation is null. The results are plotted in 4.

Table 6: RAG Pipeline Performance on Synthetic Data with 60% retrieval accuracy aveaged over 3 runs

Model	Generation Accuracy (%)	Avg Response Time (s)
Qwen	65 ± 0.0	5.89
Gemini Flash	70.0 ± 0.0	2.97
Gemini Pro	70.0 ± 0.0	3.94

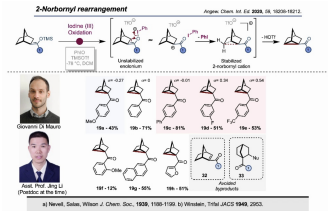
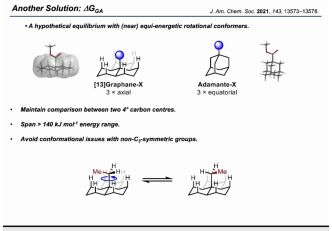
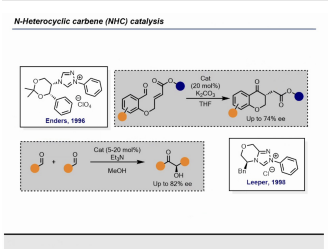
Slide	Question	Options	Ans.
	In the context of the iodine(III)-mediated oxidation shown, which of the following substituents most favors the formation of stabilized 2-norbornyl cations and avoids byproducts like compound 32?	a) 4-methoxyphenyl ($\sigma = -0.27$) b) Phenyl ($\sigma = -0.01$) c) Trifluoromethylphenyl ($\sigma = 0.54$) d) 1-naphthyl	c)
	Why is [13]Graphane-X a better model than Adamante-X for comparing axial vs. equatorial conformers in steric energy studies?	a) It maintains C-symmetry better b) It allows introduction of sp^3 carbons c) It preserves the comparison between two quaternary carbons d) It favors formation of polar intermediates	c)
	Which of the following transformations illustrates a formal benzoin reaction under NHC catalysis?	a) Leeper's annulation with Bn-substituted NHC b) The 1996 Enders' asymmetric annulation c) K_2CO_3-promoted lactone formation via NHC d) The reaction forming β-hydroxy ketones in MeOH	d)

Figure 3: Examples of questions on the synthetic advanced chemistry MCQs benchmark

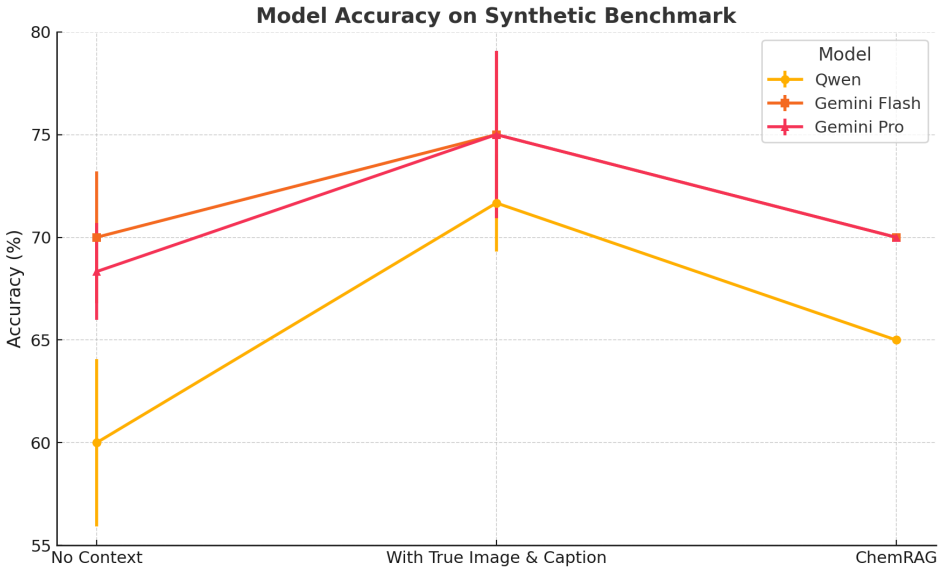


Figure 4: Accuracy and standard deviation comparison of three multimodal models (Qwen, Gemini Flash, Gemini Pro) across experimental conditions: no context, with true image and caption, and ChemRAG.

5 Interactive Web Application for Multimodal Chemistry RAG

As part of this project, we developed an interactive web application that integrates a Retrieval-Augmented Generation (RAG) system with a conversational AI interface to facilitate exploration

of chemistry lecture content. Built using Streamlit, the application enables both semantic and multimodal search across lecture slides. Users can submit queries and retrieve relevant content based on a combination of textual and visual similarity.

The retrieval pipeline relies on pre-computed FAISS indexes stored in the `saved_indexes/` directory. Textual embeddings are generated using the BGE-large-en-v1.5 model, while image captions are extracted using EasyOCR. The dataset, stored in `triplets_with_youtube.json`, consists of slide–image–caption entries aligned with corresponding YouTube timestamps. To reduce redundancy, results are grouped by speaker and spaced apart temporally.

The user interface supports interactive chat with state-of-the-art multimodal models, such as Qwen2.5-VL-7B and Gemini Pro, which can be selected via a dropdown menu. Users can ask questions about retrieved slides, and the system generates grounded responses based on both the image and its caption. Chat history is preserved across turns to support coherent multi-turn interactions.

The number of results and timestamp spacing are configurable, making the application a flexible tool for chemistry education.

The system serves as a practical demonstration of RAG in a multimodal scientific context and provides a modular testbed for benchmarking AI models under controlled conditions. Its architecture supports easy extension to other scientific disciplines and educational datasets.

The application is publicly available at: `https://chem-rag.streamlit.app`.

All indexes and image–text pairs are hosted in a private Hugging Face repository and are automatically loaded by the application. The GitHub repository includes the full pipeline that converts lecture videos into aligned slide–caption pairs. To update the indexes with new content, videos must first be processed through this pipeline and uploaded to Hugging Face; updated indexes can then be generated locally in approximately 20 minutes.

6 Conclusion

This work presents a multimodal Retrieval-Augmented Generation (RAG) system tailored to advanced chemistry education, combining state-of-the-art vision-language models with a targeted retrieval pipeline. By integrating both image and text representations, we improve the contextual understanding of domain-specific lecture slides and attempt to demonstrate the efficacy of RAG-based architectures in addressing complex multiple-choice questions grounded in real chemical content.

Our comprehensive benchmark, covering both challenging real-world data and a carefully curated synthetic set, reveals notable differences in how large multimodal models leverage contextual information. While models like Gemini Pro perform well even without explicit context, they still benefit from access to relevant slide-caption pairs, confirming the value of multimodal retrieval. The performance gap between no-context and RAG-enabled settings illustrates the importance of context in fostering chemical reasoning.

Beyond algorithmic development, we deployed a web application that integrates these components into a unified interface for exploration and interaction. This platform serves as a testbed for evaluating model performance in realistic educational workflows and supports future experimentation across diverse chemistry topics.

7 Future Work

This project opens several promising directions for further development:

- **Fine-tuning and domain adaptation:** While our system uses general-purpose multimodal models, fine-tuning them on the chemistry dataset, particularly slide-caption pairs, could yield substantial gains in answer accuracy.
- **Evaluation with stronger models:** Testing the benchmark with proprietary state-of-the-art models such as GPT-4o or Claude 3 could provide insights into the current performance ceiling and help calibrate future objectives.

- **Retrieval optimization:** Incorporating advanced retrieval mechanisms (e.g., reranking, context compression, cross-modal attention) may further reduce irrelevant content and allow better utilization of context windows in large models.
- **Multilingual and cross-disciplinary extension:** The modular architecture could be adapted for use in other scientific domains or translated content, making it a generalizable framework for multimodal education.
- **User studies and pedagogy:** Conducting controlled user studies with students and educators would help assess the pedagogical value of the system and guide interface design and feature priorities.

By bridging advanced machine learning with chemical education, this project contributes both a practical tool and a methodology for evaluating the capabilities of current and future multimodal AI systems in structured scientific reasoning.

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